

Machine Learning-Based Prediction of Stroke Severity and Functional Outcome

 **Student**
[Your Name]

 **Level**
4th Year Engineering – Data Science

 **Supervisor**
[Supervisor Name]

Polytechnic School of Sousse

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Scientific Research Project in Artificial Intelligence

Research Context & Problem Statement

Understanding Stroke (AVC)

Medical Emergency

Interrupted blood supply to the brain, leading to potential long-term disability or death.

Importance of Prediction

Early and accurate prediction of severity is crucial for personalized treatment and resource allocation.

The Research Problem

Clinical Limitations

Traditional clinical scores often lack the precision needed for complex, multi-variable patient data.

Real-world Challenges

High variability in patient outcomes and the need for rapid, data-driven decision-making in clinical settings.

Research Objectives & Questions

🎯 Research Objectives

Main Objective

Develop a robust Machine Learning framework to predict stroke severity and functional outcomes using the Tunisian AVC DataF cohort.

- ✓ Implement and compare multiple ML models.
- ✓ Optimize feature selection for clinical relevance.
- ✓ Achieve high predictive accuracy (>95%).

❓ Research Questions

- Which ML algorithms perform best on Tunisian clinical data?
- What are the most significant biological and clinical predictors of stroke outcome?
- How can ML models assist clinicians in early-stage decision making?

Literature Review / State of the Art

Comparative Analysis of Related Works

Study	Focus Area	Key Findings / Limitations
Kessler et al.	Imaging-based prediction	High accuracy but requires expensive MRI data; limited accessibility.
Sung et al.	Clinical risk factors	Effective for long-term risk; less precise for acute severity prediction.
Ntaios et al.	Stroke recurrence	Strong focus on secondary prevention rather than initial outcome.

Research Gap

- ✖ Lack of validation on North African cohorts.
- ✖ High dependency on costly imaging.

Innovation

Localized precision using the Tunisian DataF cohort with 139 clinical and biological variables.

Proposed Methodology

CRISP-DM Framework

Standard process for data mining and ML workflow.

-  Data Collection & Understanding
-  Data Preprocessing & Cleaning
-  Model Training & Optimization
-  Evaluation & Validation
-  Deployment & Prediction

Models & Technical Stack

Random Forest

Gradient Boosting

SVM

Decision Tree

Logistic Regression

 **Language:**Python

 **Libraries:**Pandas, Scikit-learn

 **Visualization:**Matplotlib, Seaborn

 **Environment:**Jupyter, Colab, VS Code

Dataset Description

875

TOTAL PATIENTS

139

CLINICAL VARIABLES

Tunisian AVC DataF Cohort

📅 2017 — 2025

Data Composition

Structured clinical and biological data including demographics, symptoms, blood markers, and medical history.

Target Variables

Primary focus on Stroke Severity (NIHSS score) and Functional Outcome (mRS score) at discharge.

Significance

One of the most comprehensive longitudinal stroke datasets in the North African region.

Data Preprocessing Pipeline

Cleaning & Engineering

1

Data Cleaning

Handling missing values via **Median Imputation** for robust statistical consistency.

2

Feature Selection

Identifying high-impact variables to reduce dimensionality and noise.

3

Label Encoding

Converting categorical clinical data into numerical formats for model compatibility.

Validation Strategy

Train/Test Split

Standard **80/20** ratio to ensure sufficient data for both learning and unbiased evaluation.

Stratified Sampling

Ensuring the distribution of stroke severity classes remains consistent across both sets.

"The preprocessing phase is critical to ensure that the ML models learn from high-quality, balanced, and representative clinical data."

Implementation & Model Development

⚙️ System Architecture

≡ Modular Pipeline

End-to-end ML pipeline designed for scalability, from data ingestion to predictive inference.

✍️ Training & Validation

Iterative training process using 5-fold cross-validation to ensure model robustness and generalizability.

⚙️ Hyperparameter Tuning

Optimization via Grid Search and Random Search to identify the most effective model configurations.

🖥️ Development Environment

The project was developed using industry-standard tools and high-performance computing environments:

 Google Colab

 VS Code

 Jupyter Notebook

 Python 3.x

🛡️ Reproducibility

Version control and environment management to ensure consistent results across different platforms.

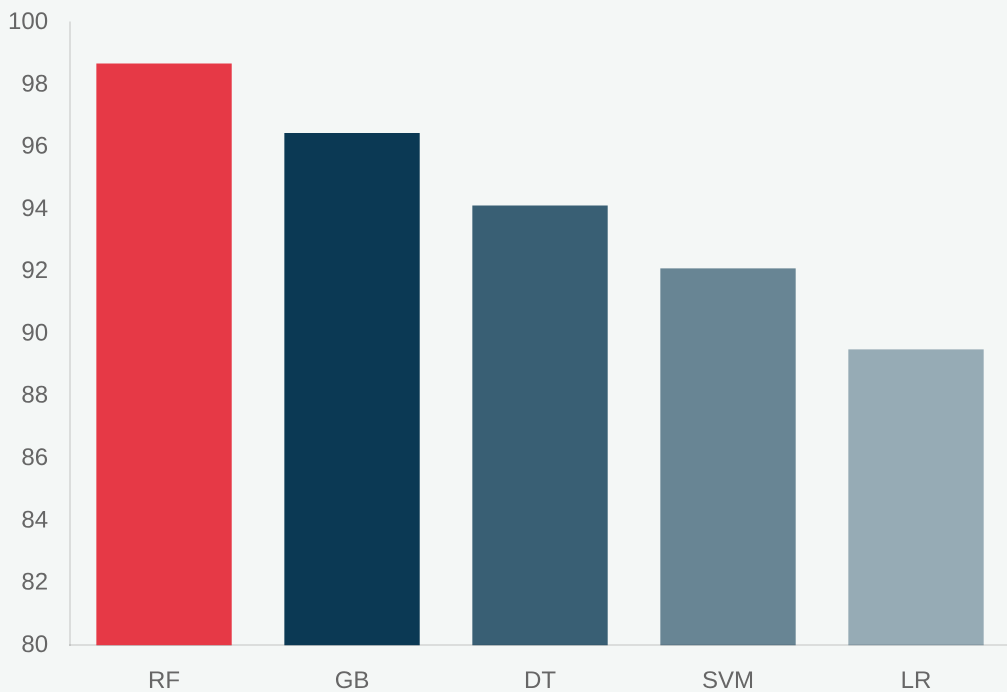
Experimental Results

Model Performance Comparison

Algorithm	Accuracy	F1-Score
Random Forest	98.68%	0.98
Gradient Boosting	96.45%	0.96
Decision Tree	94.12%	0.94
SVM	92.10%	0.91
Logistic Regression	89.50%	0.88

Metrics calculated on a 20% stratified test set from the DataF cohort.

Accuracy Visualization (%)



Discussion & Critical Analysis

Interpretation & Strengths

Model Performance

Random Forest's 98.68% accuracy suggests strong non-linear relationships within the DataF cohort variables.

Localized Value

The use of a specific Tunisian cohort ensures that the model is tailored to local clinical and biological profiles.

Feature Richness

Integrating 139 variables provides a holistic view of patient health, far exceeding traditional clinical scores.

Challenges & Effectiveness

Dimensionality Challenges

Managing 139 variables required rigorous feature selection to avoid overfitting and ensure model generalizability.

Clinical Effectiveness

The model serves as a reliable decision-support tool, enabling clinicians to prioritize high-risk patients early.

"Data-driven prognosis bridges the gap between complex biological markers and actionable clinical decisions in acute stroke care."

Conclusion & Future Work

✓ Key Contributions

Successful ML Application

Validated high-performance predictive models on the Tunisian AVC DataF cohort with 98.68% accuracy.

Clinical Impact

Identified critical biological and clinical predictors, providing a foundation for data-driven stroke management.

Localized Precision

Addressed the research gap in North African clinical data through localized model optimization.

▶▶ Future Directions



Integration of **Deep Learning (CNNs)** for automated medical imaging analysis.



Development of a **Real-time Clinical Decision Support System** for hospitals.



Expansion to **Multi-center Datasets** for broader model generalizability.



Deployment of a mobile interface for rapid bedside prognosis.